

Design and Rapid Fabrication of Integrated Fiber-Based EHD Pumps

Takafumi Morita
The University of Tokyo
Tokyo, Japan
morita@xlab.iii.u-tokyo.ac.jp

Tomoka Kurosawa
The University of Tokyo
Tokyo, Japan
kurosawa@xlab.iii.u-tokyo.ac.jp

Kanon Aoyama
The University of Tokyo
Tokyo, Japan
aoyama@xlab.iii.u-tokyo.ac.jp

Hiroki Kaimoto
The University of Tokyo
Tokyo, Japan
hkaimoto@xlab.iii.u-tokyo.ac.jp

Yu Kuwajima
Institute of Science Tokyo
Tokyo, Japan
yu.kuwajima@poliba.it

Shingo Maeda
Institute of Science Tokyo
Tokyo, Japan
maeda.s.ao@m.titech.ac.jp

Yasuaki Kakehi
The University of Tokyo
Tokyo, Japan
kakehi@iii.u-tokyo.ac.jp



Figure 1: A novel integrated fiber-based EHD pump. The pump integrates a hydraulic pump function directly into the tube itself, eliminating the need for external mechanical pumps to drive liquid flow.

Abstract

We propose a novel integrated fiber-based EHD pump for modular hydraulic control. The pump integrates a hydraulic pump function directly into the tube itself, eliminating the need for external mechanical pumps. The principle of generating liquid flow is based on the Electrohydrodynamics (EHD) phenomenon. To enable this functionality, we present a design and fabrication approach of fiber-based EHD pumps. In our proposed pump, electrodes that trigger the EHD phenomenon are embedded inside the tube using heat-shrink tubing, allowing the tube itself to silently generate liquid flow. This approach includes a design support tool that supports parametric design tailored to specific pressure and flow rate requirements for each pump, allowing for rapid and customized fabrication. This paper details the design and fabrication and demonstrates application scenarios.

CCS Concepts

• **Human-centered computing** → Interaction devices.

Keywords

Programmable Materials, Fluid Interface, Liquid Actuation, Fabrication Techniques, Electrohydrodynamics

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1 Introduction

In the field of Human-Computer Interaction (HCI), discussions around programmable matter [6, 26] have attracted attention. These aim to control the physical properties of objects, such as shape, color, and size, through programmable means. By directly manipulating the physical objects themselves, it is expected to create more intuitive and engaging interactions.

Among these, research on functional fibers that can act as sensors and actuators has been gaining attention. Due to their thin,

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flexible, and soft nature, interfaces incorporating functional fibers are expected to provide a promising new approach for seamlessly blending into daily life. While research on functional fibers with actuator capabilities is still an emerging field, they hold significant potential. To date, fibers such as McKibben-type fibers [11, 12, 14], which morph under pneumatic pressure, and fibers using twisted-then-coiled polymers [5], shape-memory alloys (SMA) [13], or liquid crystal elastomers (LCE) [4], which morph in response to heat, have been proposed. However, fibers that integrate hydraulic actuation functions are scarcely available.

Liquids, with their unique properties, enable diverse forms of expression and have been utilized in interfaces and art installations. These properties, including high thermal conductivity [7, 17, 28], high density [2, 9, 10, 20, 27], ease of coloring [8, 16, 19, 21], and the ability to engage in chemical reactions [15, 18], offer significant potential for a wide range of applications, though their use is still developing. Integrating hydraulic actuation functions into fibers could unlock new interactions by harnessing liquid behavior within the fibers, opening up previously unexplored application. However, handling liquid typically requires a complex system composed of pumps, active valves, tanks, and tubes. When these systems are integrated into interfaces, actuator noise and system configuration complexity create significant hurdles. Loud operational noise always tends to disrupt the calming and immersive experience that many liquid interfaces strive to create, and the need for complex external equipment complicates untethered operation. For these reasons, integrating hydraulic actuation functions into fibers presents a major challenge.

To address these challenges, we propose a novel integrated fiber-based EHD pump. The pump integrates a hydraulic pump function directly into the tube itself, eliminating the need for external mechanical pumps to drive liquid flow. The principle of generating liquid flow is based on the Electrohydrodynamics (EHD) phenomenon [22–24]. The EHD phenomenon occurs when electrodes are placed in a dielectric liquid and high voltage is applied, generating silent liquid flow [1, 25]. In our proposed pump, electrodes that trigger the EHD phenomenon are embedded inside the tube using heat-shrink tubing, allowing the tube itself to silently generate liquid flow.

To achieve this, we present a design and rapid fabrication approach. Additionally, a design support tool enables parametric design, accommodating the specific pressure and flow rate requirements for each pump based on user needs.

In this study, we present the design and rapid fabrication approach and show the possible application scenarios.

- (1) The design and rapid fabrication of fiber-based EHD pumps, enabling the tube itself to silently generate liquid flow.
- (2) A design support tool that allows users to enable parametric design, considering the necessary pressure and flow rate according to their specific applications.
- (3) Application scenarios utilizing modular hydraulic control with integrated fiber-based EHD pumps.

2 Design and Mechanism of the Fiber-based Pump

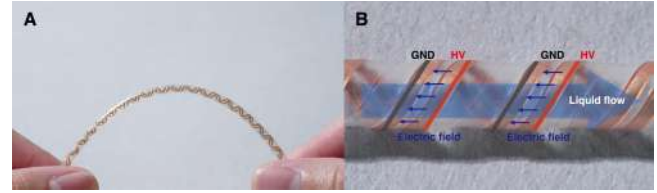


Figure 2: (A) Overall diagram of the fiber-based EHD pump. (B) Mechanism diagram of the fiber-based EHD pump.

The overall diagram is shown in Fig. 2(A). The proposed pump uses heat-shrinkable tubing as the conduit, with EHD pump electrodes attached inside the tube. The double-helix-shaped electrodes consist of two linear electrodes made from low-density polyethylene sheets and copper tape, which are processed using a cutting plotter and then spirally wound inside the tube. Connectors are attached to both ends of the tube to facilitate the connection of multiple pumps, and the ends are sealed with heat-shrinkable tubing to ensure high sealing performance. The proposed pump is thin, with a diameter in the millimeter-scale, and is soft and flexible. It has no moving parts or vibrations, allowing it to generate silent liquid flow within the tube itself.

The specific operating mechanism is shown in Fig. 2(B). The pump is filled with dielectric liquid. In this study, NOVEC7100 (3M) [3] is used as a dielectric liquid. As shown in the figure, high voltage (HV) is applied to one of the two spiral electrodes, while the other is grounded (GND), generating a high electric field between the electrodes. This electric field causes charges injected from the electrodes to move in accordance with the electric field, silently generating liquid flow through electrophoresis. The higher the electric field, the higher the pressure and flow rate of the EHD pump. Additionally, reversing the polarity of the high voltage applied to the electrodes reverses the direction of the electric field, thus directing the liquid flow in the opposite direction. To briefly summarize the principle of operation, a high voltage is applied to the pump to create a high electric field between the spiral electrodes, resulting in a liquid flow driven by the direction of the electric field.

3 Rapid Fabrication Process of Fiber-based EHD Pump

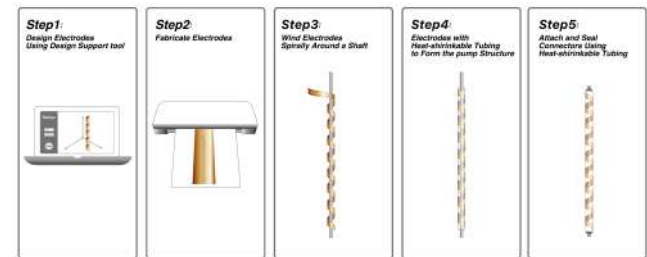


Figure 3: Overall fabrication process in five steps.

3.1 Step1: Design Electrodes Using Design Support Tool

First, the electrodes for the EHD pump are designed parametrically using the design support tool. As shown in Fig. 4(A), the design support tool allows users to design while viewing a 3D preview that reflects the selected parameter set. Once the design is finalized, pressing the export button will output a development diagram of the 3D preview as electrode data in Illustrator file format, as shown in Fig. 4(B). As the electrodes has a double-helix structure, the development diagram shows the two electrodes positioned in parallel and in close proximity. This data is then used to proceed with the fabrication of the electrodes.



Figure 4: Design electrodes using design support tool. (A) The 3D preview that corresponds to the selected parameter set. (B) The development diagram of the 3D preview as electrode data.

3.2 Step2: Fabricate Electrodes

3.2.1 Electrode fabrication process. The electrodes are created based on the development diagram data generated in Step 1. The entire electrode fabrication process is shown in Fig. 5(A) A commercially available 0.03mm-thick copper tape is cut with a cutting plotter (GRAPHTEC CE7000) while keeping the release liner intact. (B) Next, masking tape is applied over the cut copper tape that will be used as electrodes, and (C) the release liner is removed. (D) The masking tape with the attached electrodes is then adhered to a commercially available 0.06mm-thick Low Density Polyethylene (LDPE) sheet. The surface is rolled with a roller to ensure adhesion, and (E) only the masking tape is removed. (F) Finally, any excess parts of the copper tape are removed with tweezers. This process results in electrodes with cut copper tape attached on the LDPE sheet.

3.2.2 Material selection. In the electrode fabrication process described above, the selection of materials is particularly important, with the Polyethylene (LDPE) sheet being the most crucial component. The LDPE sheet has a balance of sufficient strength and flexibility, which are both significantly essential. In Step3 (described later), the electrodes fabricated in Step 2 are wound spirally around the shaft. However, due to the ductility of the metal, the copper stretches slightly during the winding process. If a less flexible sheet

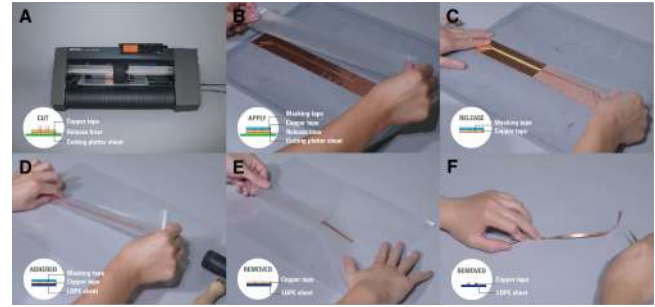


Figure 5: Electrode fabrication process: (A) copper tape is cut with a cutting plotter, (B) masking tape is applied over the cut copper tape, (C) the release liner and excess copper are removed, leaving only the electrode attached to the masking tape, (D) masking tape with electrode is adhered to an LDPE sheet, (E) masking tape is removed, (F) any small excess pieces of copper are carefully removed with tweezers, leaving only the electrode on the LDPE sheet.

material, such as PET or polyimide, is used, the copper will bend, making winding difficult. Furthermore, if the sheet lacks sufficient strength, it will not be able to maintain its shape after being wound. After testing various materials and thicknesses, we determined that a 0.06mm-thick LDPE sheet is the optimal base sheet for winding the copper electrodes and adopted it for this purpose.

3.3 Step3: Wind Electrodes Spirally Around a Shaft

In this step, the electrodes fabricated in Step 2 are wound spirally around the shaft, as illustrated in Fig. 6. The shaft used matches the pump diameter specified by the design support tool in Step 1. The electrode is wound with the copper side facing inward and the LDPE side facing outward. The winding angle is determined by the length of the terminal electrode specified in the design support tool. (A) The terminal electrode, which marks the start of the winding, is designed to align parallel to the shaft. Therefore, when winding the electrode around the shaft, the terminal electrode is first aligned and fixed parallel to the shaft, and (B) then the remaining electrode is wound around the shaft. This method ensures that the spirally wound electrodes remain in place without shifting, preventing bulkiness or gaps and allowing for easy, rapid, and precise winding.

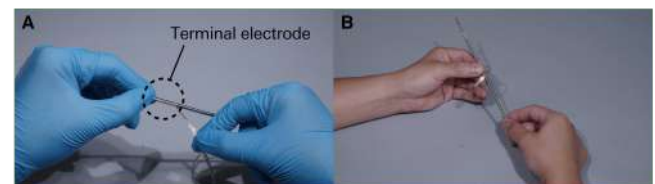


Figure 6: Winding Electrode Process. (A) The terminal electrode is first aligned and fixed parallel to the shaft. (B) The electrode is then wound spirally around the shaft.

3.4 Step4: Encase Electrodes with Heat-shrinkable Tubing to Form the Pump Structure

In this step, a heat-shrinkable tube is used to form the shape of the fiber-based pump. (A) The heat-shrinkable tube, which is slightly larger in diameter than the shaft around which the electrodes were spirally wound in Step 3, is placed over the shaft. (B) The tube is then heated from the outside with a heat gun, causing it to contract and adhere tightly to the electrodes inside, thereby forming the tube shape as illustrated in Fig. 7. (C) Once the heating is complete, the shaft is removed, and the fiber-based pump is completed.



Figure 7: Encase electrodes with heat-shrinkable tubing. (A) The heat-shrinkable tube is placed over the shaft. (B) The tube is heated from the outside. (C) The shaft is removed.

3.5 Step5: Attach and Seal Connectors Using Heat-shrinkable Tubing

In Step 4, the fiber-based pump is nearly complete, but the connectors still need to be attached to both ends of the pump. Commercially available I-type connectors are used. These connectors are inserted into each end of the pump, and the heat-shrinkable tubing that covers the connector area is heated.

4 Demonstrating Application Scenarios



Figure 8: (A) A linear serpentine actuator for bending soft elastomer actuator. (B) A planar raft actuator to propel the raft forward across the liquid surface. (C) A quiet and portable tumbler-wound indicator.

4.1 Linear Serpentine Actuator for Bio-Inspired Robotics

As shown in Fig. 8 (A), the linear serpentine actuator for bio-inspired robotics is introduced. Pressure is an important factor when using liquid flow as the driving force for soft actuators. In this demonstration, the bending motion of a soft actuator made from elastomer is shown, varying with the number of pumps connected in series. Additionally, since the pump operates quietly throughout the entire system, it is highly suitable as an actuation method for bio-inspired robotic applications.

4.2 Planar Raft Actuator for Liquid Surface Movement

We introduce a planar raft actuator designed for liquid surface movement, as shown in Fig. 8 (B). In a tube of the same diameter, flow rate is a critical factor in determining the speed of liquid movement. Here, several pumps connected in parallel are positioned beneath a raft-like structure floating on the liquid surface. By connecting them in parallel, the increased flow rate acts as the driving force to propel the raft forward. This system is self-contained and can be controlled wirelessly. Its silent operation allows it to function as an ambient actuator, gliding smoothly across the water's surface.

4.3 Tumbler-Wound Indicator for Existing Object Integration

In Fig. 8 (C), the tumbler-wound indicator exemplifies the seamless integration of the modular fiber-based pump into existing objects. Its soft and flexible nature allows it to be easily incorporated, with a tube wound around a tumbler serving as a functional indicator. This system is also self-contained, making the indicator highly portable. Since it operates without producing noise, it functions as a quiet indicator that blends seamlessly into its environment.

5 Conclusion

In this paper, we propose a novel integrated fiber-based EHD pump. The pump integrates the hydraulic pump function directly into the tube itself, eliminating the need for external mechanical pumps to drive liquid flow. To enable this functionality, we present a design and fabrication approach of fiber-based EHD pump. This approach includes a design support tool that supports parametric design tailored to specific pressure and flow rate requirements for each pump, allowing for rapid and customized fabrication. Furthermore, we explored application scenarios. With these contributions, we hope to pave the way for fibers that integrate hydraulic actuation functions, positioning them as a promising technology for seamlessly blending into everyday environments in the near future.

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